Simplifying a Derivative

MTH 207 · for §4.3 and §4.5 · factor out the **smaller** exponent

The Product, Quotient and Chain Rules hand you a *sum of terms*. Finding critical numbers and inflection points means setting that sum equal to zero — and you cannot read the zeros off a sum. You have to **factor** it into one product over one denominator. That factoring is the hard part of almost every §4.3 and §4.5 problem, and it is the only thing this sheet is about.

You already know the rule

Factor x^2 out of $x^5 + x^2$:

$$x^5 + x^2 \;=\; x^2\left(x^3 + 1\right)$$

You kept the **smaller** power and **subtracted it from each exponent**: $5 - 2 = 3$, and $2 - 2 = 0$ (and $x^0 = 1$, which is the $+1$).

That is the whole method. Nothing changes when the exponents are fractions or negative — only the arithmetic gets slightly harder.

The rule, in general

$$\text{factoring } u^m \text{ out of } u^n \text{ leaves } u^{\,n-m}$$

Why the *smaller* exponent?

Because then every leftover exponent $n - m$ is ≥ 0, so **nothing is left in a denominator and nothing is left under a radical**. That is the entire reason. If you factor out the larger one instead, the leftovers come out negative — and negative exponents in a denominator are exactly what you were trying to clear.

Recipe. Write down the exponents that appear. Circle the smaller one. Subtract it from each. Those differences are what is left inside the bracket.

Free self-check. The term you factored *out* must leave $u^0 = 1$ behind. So **one of the terms in the bracket is always a bare number** — if neither is, you factored out the wrong one, or subtracted wrong.

The bookkeeping, done slowly

Suppose a derivative comes out as

$$(6-x)^{1/2} \;-\; \tfrac{x}{2}\,(6-x)^{-1/2}$$

The exponents present are $\frac{1}{2}$ and $-\frac{1}{2}$. The smaller is $-\frac{1}{2}$, so factor $(6-x)^{-1/2}$ out. Subtract $-\frac{1}{2}$ from each exponent:

$$\underbrace{\tfrac{1}{2} - \left(-\tfrac{1}{2}\right) = 1}_{\text{first term leaves } (6-x)^1} \qquad \underbrace{-\tfrac{1}{2} - \left(-\tfrac{1}{2}\right) = 0}_{\text{second leaves } (6-x)^0 = 1}$$

so

$$(6-x)^{1/2} - \tfrac{x}{2}\,(6-x)^{-1/2} \;=\; (6-x)^{-1/2}\left[(6-x) - \tfrac{x}{2}\right]$$

Subtracting a negative is the step that goes wrong. $\frac{1}{2} - \left(-\frac{1}{2}\right)$ is $\frac{1}{2} + \frac{1}{2} = 1$, not 0.

Example 1 $\quad f(x) = x\sqrt{6-x}$

Domain: $6 - x \geq 0$, so $x \leq 6$.

First derivative — Product Rule, then Chain Rule

$$f(x) = x\,(6-x)^{1/2}$$ — write the radical as a power first

$$f'(x) = 1\cdot(6-x)^{1/2} + x\cdot\tfrac{1}{2}(6-x)^{-1/2}\cdot(-1)$$ — Product Rule; the (-1) is the Chain Rule

$$= (6-x)^{1/2} - \tfrac{x}{2}(6-x)^{-1/2}$$ — tidy the second term

$$= (6-x)^{-1/2}\left[(6-x) - \tfrac{x}{2}\right]$$ — **factor out** $(6-x)^{-1/2}$, the smaller exponent

$$= (6-x)^{-1/2}\left[6 - \tfrac{3x}{2}\right]$$ — combine the x terms: $-x - \tfrac{x}{2} = -\tfrac{3x}{2}$

$$= \frac{12-3x}{2\sqrt{6-x}} = \frac{3(4-x)}{2\sqrt{6-x}}$$ — multiply top and bottom by 2, then factor 3

Critical numbers: the numerator is 0 at $\mathbf{x = 4}$, and f' does not exist at $\mathbf{x = 6}$ (the denominator is 0 there, and $x = 6$ *is* in the domain).

Second derivative — same method, one step further out

$$f'(x) = \tfrac{3}{2}(4-x)(6-x)^{-1/2}$$ — rewrite before differentiating again

$$f''(x) = \tfrac{3}{2}\left[(-1)(6-x)^{-1/2} + (4-x)\left(-\tfrac{1}{2}\right)(6-x)^{-3/2}(-1)\right]$$ — Product Rule, Chain Rule again

$$= \tfrac{3}{2}\left[-(6-x)^{-1/2} + \tfrac{1}{2}(4-x)(6-x)^{-3/2}\right]$$ — the two minus signs make a plus

$$= \tfrac{3}{2}(6-x)^{-3/2}\left[-(6-x) + \tfrac{1}{2}(4-x)\right]$$ — **smaller exponent is** $-\tfrac{3}{2}$; $-\tfrac{1}{2} - \left(-\tfrac{3}{2}\right) = 1$

$$= \tfrac{3}{2}(6-x)^{-3/2}\left[-6 + x + 2 - \tfrac{x}{2}\right]$$ — expand the bracket

$$= \tfrac{3}{2}(6-x)^{-3/2}\cdot\frac{x-8}{2} = \frac{3(x-8)}{4(6-x)^{3/2}}$$

Read the domain before you answer. $f'' = 0$ at $x = 8$ — but 8 is *not in the domain*, so there is **no inflection point**. For every $x < 6$ the top is negative and the bottom is positive, so $f'' < 0$ and f is **concave down on its whole domain**.

Example 2 $f(x) = 5x^{2/3} - 2x^{5/3}$

$$f'(x) = 5\cdot\tfrac{2}{3}x^{-1/3} - 2\cdot\tfrac{5}{3}x^{2/3} = \tfrac{10}{3}x^{-1/3} - \tfrac{10}{3}x^{2/3}$$ — Power Rule: subtract 1 from each exponent

$$= \tfrac{10}{3}x^{-1/3}[1-x]$$ — **smaller is** $-\tfrac{1}{3}$; $\tfrac{2}{3} - \left(-\tfrac{1}{3}\right) = 1$

$$= \frac{10(1-x)}{3\sqrt[3]{x}}$$

Critical numbers: $\mathbf{x = 1}$ (numerator zero) and $\mathbf{x = 0}$ (f' does not exist, and $f(0) = 0$ is defined, so 0 counts).

$$f''(x) = \tfrac{10}{3}\left(-\tfrac{1}{3}\right)x^{-4/3} - \tfrac{10}{3}\cdot\tfrac{2}{3}x^{-1/3} = -\tfrac{10}{9}x^{-4/3} - \tfrac{20}{9}x^{-1/3}$$

$$= -\tfrac{10}{9}x^{-4/3}[1+2x]$$ — **smaller is** $-\tfrac{4}{3}$; $-\tfrac{1}{3} - \left(-\tfrac{4}{3}\right) = 1$

$$= \frac{-10(1+2x)}{9\,x^{4/3}}$$

Example 3 $f(x) = \dfrac{x}{\sqrt{x+2}}$

Domain: $x+2>0$, so $x>-2$ — strict, because it is a denominator.

$$f(x) = x\,(x+2)^{-1/2}$$ a denominator is a negative exponent

$$f'(x) = (x+2)^{-1/2} + x\left(-\tfrac{1}{2}\right)(x+2)^{-3/2}$$ Product Rule

$$= (x+2)^{-3/2}\left[(x+2) - \tfrac{x}{2}\right]$$ **smaller is** $-\frac{3}{2}$; $-\frac{1}{2} - \left(-\frac{3}{2}\right) = 1$

$$= (x+2)^{-3/2}\left[\tfrac{x}{2}+2\right] = \frac{x+4}{2\,(x+2)^{3/2}}$$

The same trap again. $f'=0$ at $x=-4$, which is *not* in the domain $x>-2$. So f has **no critical numbers at all**, and since $x+4>0$ throughout the domain, f is **increasing everywhere it exists**.

Now you try

Same method, different numbers. Answers are given so you can check yourself — the *work* is the point, so do it before you look.

1. $f(x) = x\sqrt{8-x}$ $\qquad f'(x) = \dfrac{16-3x}{2\sqrt{8-x}}, \quad f''(x) = \dfrac{3x-32}{4\,(8-x)^{3/2}}$

2. $f(x) = 4x^{3/2} - x^{5/2}$ $\qquad f'(x) = \dfrac{\sqrt{x}\,(12-5x)}{2}, \quad f''(x) = \dfrac{3\,(4-5x)}{4\sqrt{x}}$

3. $f(x) = \dfrac{x}{\sqrt{x+5}}$ $\qquad f'(x) = \dfrac{x+10}{2\,(x+5)^{3/2}}, \quad f''(x) = \dfrac{-\,(x+20)}{4\,(x+5)^{5/2}}$

For each one, also ask the question the answers do not give you: *which of those zeros are actually in the domain?* In (1) the domain is $x\le 8$ and $\frac{16}{3}\approx 5.3$ is in it; in (3) the domain is $x>-5$ and $x=-10$ is not.

Three habits that fix most of the lost marks. (1) Write every radical and every denominator as a *power* before you differentiate. (2) Factor out the smaller exponent, and check that one bracket came out a bare number. (3) Compare your zeros against the *domain* before you call anything a critical number or an inflection point.